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1 考点 1: 投资组合构建

1. An increase in which of the following factors will expand the no-trade region for the alpha of an asset? I. Risk aversion. II. Marginal contribution to active risk.
- A. I only.
- B. II only.
- C. Both I and II.
- D. Neither I nor II.

答案:C

解析:

- A 选项错误。** 只有风险厌恶度增加会扩大非交易区间，但题干还涉及第二项。
- B 选项错误。** 只有积极风险的边际贡献增加也会扩大非交易区间，但题干还涉及第一项。
- C 选项正确。** 风险厌恶度和积极风险的边际贡献增加都会使非交易区间变大，二者均成立。
- D 选项错误。** 至少有一项成立，不能选 Neither。

考察: 非交易区间影响因素

2. An analyst has forecasts for stocks X, Y, and Z, but not for stocks U and V. Stocks X, Y, and U are part of the benchmark portfolio, while stocks Z and V are not. Which of the following is correct regarding the specific weights the analyst should assign when constructing a portfolio to track this benchmark?
- A. $w_Z = 0$.
- B. $w_U = 0$.
- C. $w_Z = (w_X + w_Y)/2$.
- D. $w_Z = w_U = w_V$.

答案:A

解析:

- A 选项正确。** 对于有预测但不在基准中的股票 Z，其在跟踪投资组合中的权重应设为 0。
- B 选项错误。** 股票 U 在基准中，其权重应根据其他资产的 alpha 确定，不应为 0。
- C 选项错误。** 将权重设为其他股票的平均值没有理论依据。
- D 选项错误。** 不同股票的权重应根据其在基准中的位置和预测情况分别确定，不能简单地设为相等。

考察: 基准外资产的权重处理

3. During the 2007–2008 financial crisis, concerns arose about the high interconnectedness of institutions like Bear Stearns and Lehman Brothers with other financial firms. What can such interdependency among financial institutions lead to?
- A. Negative kurtosis.
- B. Lower variance of returns.
- C. Fat-tailed distributions.
- D. Excess system-wide leverage.

答案:C

解析:

- A 选项错误。** 互联性增强通常导致厚尾分布（正峰度），而非负峰度。
- B 选项错误。** 金融机构高度关联会增加系统风险，通常不会降低收益方差。
- C 选项正确。** 金融机构间的高度依赖会导致收益分布出现厚尾，极端风险事件概率上升。
- D 选项错误。** 虽然杠杆可能增加，但互联性本身主要导致风险分布厚尾，而非直接导致杠杆过高。

考察: 金融机构互联性与厚尾风险

4. An analyst regresses the returns of 100 stocks against a major market index. The resulting 100 alphas have a residual risk of 20% and an information coefficient of 10%. If the alphas are normally distributed with a mean of 0%, approximately how many stocks have an alpha greater than 4% or less than -4%?

- A. 5
- B. 10
- C. 20
- D. 25

答案:A

解析:

- A 选项正确。** α 的标准差为 $0.2 \times 0.1 = 0.02$ ，4% 为 2 倍标准差，正态分布下约有 5% 的样本超出区间，100 只股票约有 5 只。
- B 选项错误。** 10 只股票超出 2 倍标准差的概率远高于正态分布理论值。
- C 选项错误。** 20 只股票超出 2 倍标准差的概率明显过高。
- D 选项错误。** 25 只股票超出 2 倍标准差的概率远超正态分布理论。

考察: α 分布与标准差概率

2 考点 2: 投资组合风险：分析方法

1. Given the following portfolio data, which recommendation would best improve the risk/return tradeoff of the portfolio?

Asset	Expected return	Std. Dev.	Current Weight	Marginal Return	Marginal Risk	Marginal Return / Marginal Risk
Asset 1	7.10%	17.00%	38.70%	3.10%	13.99%	22.17%
Asset 2	8.00%	40.60%	6.20%	4.00%	23.93%	16.71%
Asset 3	6.70%	44.80%	5.50%	2.70%	23.39%	11.55%
Asset 4	6.90%	21.40%	14.60%	2.90%	13.86%	20.92%
Risk free	4.00%	0.00%	35.00%			

- A. Increase allocations to assets 1 and 3, decrease allocations to assets 2 and 4.
- B. Increase allocations to assets 1 and 2, decrease allocations to assets 3 and 4.
- C. Increase allocations to assets 2 and 3, decrease allocations to assets 1 and 4.
- D. Increase allocations to assets 1 and 4, decrease allocations to assets 2 and 3.

答案:D

解析:

- A 选项错误。** 资产 3 的边际收益/边际风险比最低，应减少配置。
- B 选项错误。** 资产 4 的边际收益/边际风险比最高，应增加配置。
- C 选项错误。** 资产 4 应增加配置，资产 3 应减少配置。
- D 选项正确。** 优化风险收益权衡应增加边际收益/边际风险比高的资产（1 和 4），减少比值低的资产（2 和 3）。
- 考察:** 边际收益/风险比优化配置

2. A risk manager is assessing the risk profile of a stock portfolio valued at JPY 150 billion, which includes JPY 30 billion in stock ABC. The annual standard deviation of returns for stock ABC is 12%, and for the overall portfolio is 20%. The correlation between stock ABC and the portfolio is 0.5. Assuming a 1-year 95% VaR and normal return distribution, what is the estimated component VaR of stock ABC?

- A. JPY 2.958 billion
- B. JPY 3.546 billion
- C. JPY 4.734 billion
- D. JPY 6.120 billion

答案:A

解析:

- A 选项正确。** 先计算 ABC 的 VaR： $30 \times 0.12 \times 1.645 = 5.922$ ，再乘以相关系数 $0.5 \times 5.922 = 2.961$ ，最接近 2.958。
- B 选项错误。** 3.546 高于实际 component VaR 计算结果。
- C 选项错误。** 4.734 远高于实际 component VaR。
- D 选项错误。** 6.120 明显高于正确答案。
- 考察:** 成分 VaR 计算方法

3. Given the following information, what is the percentage contribution to VaR from Asset X? There are two assets in a portfolio: Asset X marginal VaR: 0.06250, Asset X value: \$8,000,000 Asset Y marginal VaR: 0.15000, Asset Y value: \$5,000,000

- A. 40.00%
- B. 25.00%
- C. 51.06%
- D. 48.94%

答案:D

解析:

- A 选项错误。** 40.00% 不是根据 component VaR 计算得出的正确比例。
- B 选项错误。** 25.00% 是 marginal VaR 权重，不是贡献比例。

C 选项错误。 51.06% 是 Asset Y 的贡献比例。
D 选项正确。 Asset X 的 component VaR 为 $0.0625 \times 8,000,000 = 500,000$, Asset Y 为 $0.15 \times 5,000,000 = 750,000$, X 的贡献比例为 $500,000/(500,000 + 750,000) = 48.94\%$ 。
考察: 成分 VaR 贡献度计算

4. A portfolio consists of two securities with the following characteristics: Investment in A: USD 2.0 million Investment in B: USD 3.0 million Volatility of A: 10% Volatility of B: 5% Correlation between A and B: 20% What is the closest value to the diversified portfolio VaR at the 95% confidence level?

A. 1
B. \$18,400
C. \$245,000
D. \$495,000

答案:D
解析:
A 选项错误。 1 不是 VaR 的合理数值。
B 选项错误。 \$18,400 远低于实际 VaR 结果。
C 选项错误。 \$245,000 低于实际组合 VaR。
D 选项正确。 组合方差 = $(0.4^2 \times 0.1^2) + (0.6^2 \times 0.05^2) + 2 \times 0.4 \times 0.6 \times 0.1 \times 0.05 \times 0.2 = 0.0016 + 0.0009 + 0.00024 = 0.00274$, 标准差 $\approx 5.24\%$, VaR = $1.65 \times 0.0524 \times \$5m \approx \$432,300$, 最接近 \$495,000。
考察: 投资组合分散化 VaR 计算

5. A portfolio manager estimates the VaRs for two positions as follows: VaR(A) = \$6.0 million, VaR(B) = \$3.0 million. What is the portfolio VaR if the returns of the two securities are uncorrelated, and what is the portfolio VaR if the returns are perfectly correlated?

A. For zero correlation, VaR is \$6.71 million; for perfect correlation, VaR is \$9.00 million.
B. For zero correlation, VaR is \$5.00 million; for perfect correlation, VaR is \$7.00 million.
C. For zero correlation, VaR is \$9.00 million; for perfect correlation, VaR is \$3.00 million.
D. For zero correlation, VaR is \$7.71 million; for perfect correlation, VaR is \$12.00 million.

答案:A
解析:
A 选项正确。 零相关时组合 VaR 为 $\sqrt{6^2 + 3^2} = 6.71$ 百万, 完全相关时为 $6 + 3 = 9$ 百万。
B 选项错误。 5 和 7 百万均不是正确的 VaR 计算结果。
C 选项错误。 9 百万是完全相关时的 VaR, 3 百万不是零相关时的 VaR。
D 选项错误。 7.71 和 12 百万均高于实际计算结果。
考察: 相关性对组合 VaR 的影响

6. Pension analysts at a large firm use a two-step process for asset and risk management: first, they allocate assets across four classes using a VaR-based risk budget; second, they set a maximum tracking error for each class and assign active risk budgets to managers. Given the table below, which statements are correct? I. Using VaR as the risk budgeting measure, the emerging markets class has the smallest risk budget. II. If an additional dollar is invested, the marginal impact on portfolio VaR is greatest for small caps. III. Lowering the maximum tracking error allowance gives managers more freedom to achieve excess returns. IV. Setting strict risk limits and monitoring guarantees that risk limits will not be exceeded.

	Return (%)	Volatility (%)	Allocation (%)	VAR (USD)	Marginal VAR
Small Cap	0.20	2.66	35.0	6,491	0.055
Large Cap	0.15	2.33	40.0	6,497	0.044
Commodities	0.10	1.91	16.7	2,216	0.020
Emerging markets	0.15	2.70	8.3	1,570	0.047
Total VAR: 13,322					

A. I and II only

B. I, II, III, and IV

C. II and III

D. I only

答案:A

解析:

A 选项正确。 I 正确：新兴市场的 VAR 最小，风险预算最小。II 正确：小盘股的边际 VAR 最大，增加投资对组合 VAR 影响最大。

B 选项错误。 III 错误：降低跟踪误差限制会减少而非增加经理的操作自由。IV 错误：即使有严格风险限额，也不能保证风险绝不超限。

C 选项错误。 只有 II 正确，III 错误。

D 选项错误。 只有 I 正确，II 也应包括。

考察: VAR 风险预算与边际风险

7. Given the following information, what is the percentage contribution to VaR from Asset P? A portfolio has two assets: P and Q. Asset P marginal VaR: 0.04, Asset P value: \$15,000,000 Asset Q marginal VaR: 0.10, Asset Q value: \$5,000,000

A. 45.46%

B. 28.57%

C. 75.00%

D. 54.54%

答案:D

解析:

A 选项错误。 45.46% 是资产 Q 的 VaR 贡献比例。

B 选项错误。 28.57% 是资产 P 的边际 VaR 权重，不是其 VaR 贡献比例。

C 选项错误。 75.00% 是资产 P 的资产价值权重，不是其 VaR 贡献比例。

D 选项正确。 资产 P 的成分 VaR 为 $0.04 \times 15,000,000 = 600,000$ ，资产 Q 的成分 VaR 为 $0.10 \times 5,000,000 = 500,000$ 。资产 P 的贡献比例为 $600,000 / (600,000 + 500,000) = 54.54\%$ 。

考察: 成分 VaR 贡献度计算

3 考点 3: 投资管理中的 VaR 和风险预算编制

1. The Q-statistic is a useful measure for assessing liquidity risk in hedge funds. Which of the following statements about the statistical significance of the Q-statistic is true?
- A. Smaller p-values indicate that autocorrelations are more statistically significant.
 - B. A p-value of 0.1 means we are 99% confident in rejecting the null hypothesis of no correlation.
 - C. We can reject the null hypothesis of positive autocorrelations when each lagged autocorrelation is close to zero.
 - D. Larger p-values suggest that autocorrelations are more statistically significant.

答案:A
解析:
A 选项正确。 p 值越小, 拒绝原假设 (自相关为零) 的信心越足, 说明自相关性越显著。
B 选项错误。 p 值为 0.1 表示我们有 90% 的信心拒绝原假设, 而不是 99
C 选项错误。 原假设通常是自相关为零, 不是正自相关。
D 选项错误。 p 值越大, 说明自相关性越不显著。
考察: Q 统计量与 p 值显著性

2. A corporate sponsor manages a defined benefit pension plan and must consider funding risk. Which of the following statements about the plan’s funding risk is correct?
- A. As the horizon for expected payouts lengthens, funding risk decreases.
 - B. A decline in interest rates will mitigate funding risk.
 - C. The funding risk is effectively borne by the employees.
 - D. For the plan sponsor, funding risk is the genuine long-term risk.

答案:D
解析:
A 选项错误。 支付期限越长, 资产与负债的不匹配风险可能越大, 资金风险可能增加。
B 选项错误。 利率下降会提高负债的现值, 通常会加剧而非减少资金风险。
C 选项错误。 在固定收益计划中, 资金风险由发起人承担, 而非雇员。
D 选项正确。 资金风险代表了资产和负债的长期不匹配, 是计划发起人面临的真正长期风险。
考察: 固定收益计划的资金风险

3. An analyst at Quantum Capital needs to invest a small amount of new capital into a fund benchmarked against an index. Instead of adding a new asset, the analyst will increase the holding of one of the four assets in the table below. The analyst wants to select the asset with the lowest marginal VaR, provided its Treynor ratio is at least 0.08. Assuming a risk-free rate of 3%, which asset should be selected?

Asset	Portfolio Weight	Expected Return	Beta to Index	Beta to Portfolio
W	1.5%	15%	1.5	1.1
X	1.1%	11%	0.9	0.8
Y	0.9%	9%	0.8	0.7
Z	0.5%	8%	0.5	1.0

- A. Asset W
- B. Asset X
- C. Asset Y
- D. Asset Z

答案:B
解析:
A 选项错误。 资产 W 的特雷诺比率为 $(15\% - 3\%) / 1.5 = 0.08$, 符合条件, 但其组合 Beta 为 1.1, 高于资产 X。
B 选项正确。 资产 X 的特雷诺比率为 $(11\% - 3\%) / 0.9 = 0.089$, 符合条件, 其组合 Beta 为 0.8, 是所有符合条件资产中最低的。

C 选项错误。资产 Y 的特雷诺比率为 $(9\% - 3\%) / 0.8 = 0.075$ ，低于 0.08，不符合条件。
D 选项错误。资产 Z 的特雷诺比率为 $(8\% - 3\%) / 0.5 = 0.1$ ，符合条件，但其组合 Beta 为 1.0，高于资产 X。
考察: 特雷诺比率与边际风险决策

4. For a portfolio of illiquid assets, analysts may smooth returns by adjusting valuations. Which of the following is NOT a typical consequence of this practice?
- A. A higher reported Sharpe ratio.
 - B. Lower reported volatility.
 - C. Higher serial correlation in returns.
 - D. A higher measured market beta.

答案:D
解析:

A 选项错误。回报平滑会降低报告的波动率，从而人为地提高夏普比率。
B 选项错误。回报平滑的直接结果就是降低报告的波动率。
C 选项错误。回报平滑会导致一个时期的回报部分地计入下一个时期，从而产生正的序列相关性。
D 选项正确。回报平滑会降低投资组合回报与市场回报的协方差，从而导致市场贝塔系数被低估，而不是高估。
考察: 回报平滑的后果

5. Risk management for hedge funds presents unique challenges compared to traditional investment firms. Which of the following statements about hedge fund risk management are correct? I. Due to long/short positions, derivatives, and leverage, hedge fund exposures can change rapidly, making monthly returns insufficient for risk assessment. II. Hedge funds often use OTC derivatives and hold illiquid assets, causing their returns to exhibit lower serial correlation than mutual fund returns. III. Strategies that use leverage and rely on quick exits require careful monitoring and management of liquidity risk. IV. Hedge fund returns can resemble a basket of exotic derivatives with nonlinear payoffs, so risk assessment based on past performance can be misleading.
- A. I, II, III, and IV
 - B. I, III, and IV
 - C. I and III
 - D. II and IV

答案:B
解析:

A 选项错误。包含错误的陈述 II。
B 选项正确。陈述 I、III 和 IV 都正确描述了对冲基金风险管理的挑战。陈述 II 错误，因为持有非流动性资产通常会导致更高的序列相关性，而不是更低。
C 选项错误。遗漏了正确的陈述 IV。
D 选项错误。包含错误的陈述 II，并遗漏了正确的陈述 I 和 III。
考察: 对冲基金风险管理挑战

6. The Zenith Corp pension fund has \$20 billion invested in equities. Assume a normal distribution and a volatility of 18% per annum. The fund measures absolute risk with a 95%, one-year VAR, which is \$4.5 billion. The pension plan wants to allocate this risk to two equity managers, each with the same VAR budget. Given that the correlation between the managers is 0.2, the VAR budget for each should be closest to:
- A. \$4.5 billion
 - B. \$3.5 billion
 - C. \$2.9 billion
 - D. \$2.25 billion

答案:C

解析:

A 选项错误。 \$4.5 亿是总 VaR，不是每个经理的预算。

B 选项错误。 \$3.5 亿不是根据 VaR 分配公式计算出的正确结果。

C 选项正确。 设每个经理的 VaR 预算为 x。 则组合 VaR 的计算公式为：

$$\begin{aligned}\text{VaR}_{total}^2 &= x^2 + x^2 + 2\rho x^2 \\ \$4.5^2 &= 2x^2 + 2(0.2)x^2 = 2.4x^2 \\ x &= \sqrt{\frac{\$4.5^2}{2.4}} = \sqrt{\frac{20.25}{2.4}} \approx \$2.9 \text{ billion}\end{aligned}$$

D 选项错误。 \$2.25 亿是总 VaR 的一半，只有在相关性为 1 时才成立。

考察: VaR 风险预算分配

7. StarLink Satellites has a defined benefit pension scheme with assets of \$200 million and liabilities of \$180 million. The annual growth of liabilities is expected to be 5.0% with 3.0% volatility. The annual return on pension assets has an expected value of 8.0% with 15% volatility. The correlation between asset return and liability growth is 0.40. What is the 95% surplus at risk for StarLink?

- A. \$15.51 million
- B. \$28.28 million
- C. \$39.51 million
- D. \$46.51 million

答案:C

解析:

A 选项错误。 \$15.51 百万不是根据风险敞口计算得出的正确盈余风险值。

B 选项错误。 \$28.28 百万是盈余的标准差，不是最终的风险敞口。

C 选项正确。 首先，计算预期盈余增长：\$200M * 8% - \$180M * 5% = \$7M。 其次，计算盈余波动性（标准差）：

$$\begin{aligned}\text{Var}(\text{Surplus}) &= (200^2 \times 0.15^2) + (180^2 \times 0.03^2) - 2 \times 0.4 \times (200 \times 0.15) \times (180 \times 0.03) = 799.56 \\ \text{StDev}(\text{Surplus}) &= \sqrt{799.56} = \$28.28\text{M}\end{aligned}$$

最后，计算 95% 置信水平下的风险敞口：

$$\text{Surplus at Risk} = 1.645 \times \$28.28\text{M} - \$7\text{M} = \$46.51\text{M} - \$7\text{M} = \$39.51\text{M}$$

D 选项错误。 \$46.51 百万是盈余的 VaR，但没有减去预期盈余增长。

考察: 养老金盈余风险计算

8. A hedge fund has a long position of USD 400 million and a short position of USD 250 million. The fund’ s equity is USD 200 million, and its overall beta is 0.8. Calculate the Gross and Net leverage.

- A. 3.25 and 0.75
- B. 3.25 and 1.25
- C. 0.75 and 3.25
- D. 1.25 and 0.75

答案:A

解析:

A 选项正确。 总杠杆 = (多头头寸 + 空头头寸) / 净值 = (\$400M + \$250M) / \$200M = 3.25。 净杠杆 = (多头头寸 - 空头头寸) / 净值 = (\$400M - \$250M) / \$200M = 0.75。

B 选项错误。 净杠杆计算错误。

C 选项错误。 总杠杆和净杠杆的值混淆了。

D 选项错误。 总杠杆计算错误。

考察: 对冲基金杠杆率计算

9. Hedge fund manager Sarah Chen has managed a portfolio for over a decade. During the market turmoil of 2020, she was surprised to see her portfolio’s returns fall sharply with the market, despite the fund being structured to have a beta uncorrelated with the market index. She considers several concepts to explain this. Which of the following risk-related items could NOT be used to explain this observation?

- A. Nonlinearity risk.
- B. Systemic correlation.
- C. Asymmetric correlation.
- D. Phase-locking behavior.

答案:B

解析:

- A 选项错误。**非线性风险（如期权类敞口）在市场大跌时可能被触发，是合理的解释。
- B 选项正确。**“系统性相关性”不是一个标准的金融术语。虽然系统性风险会导致相关性增加，但这个词本身并不成立。
- C 选项错误。**非对称相关性指的是在市场下跌时相关性会显著增加，这是一个有效的解释。
- D 选项错误。**锁相行为描述了在危机事件中，通常不相关的资产会变得高度相关，这也是一个合理的解释。

考察: 危机中相关性的变化

10. An analyst provides the following information to a pension fund advisor. The fund has an initial surplus of USD 10 million. To evaluate the surplus adequacy, the advisor estimates the surplus value at the end of one year, assuming returns and liability growth are jointly normal with a correlation of 0.7. With a 95% confidence level, what will the surplus value be greater than or equal to?

	Pension Assets	Pension Liabilities
Amount (in USD million)	120	110
Expected Annual Growth	5%	6%
Annual Volatility of Growth	12%	4%

- A. USD -11.8 million
- B. USD -9.9 million
- C. USD -4.5 million
- D. USD 0 million

答案:B

解析:

- A 选项错误。**计算结果不符。
- B 选项正确。**首先，计算预期期末盈余：

$$\text{Expected Surplus} = (120 \times 1.05) - (110 \times 1.06) = 126 - 116.6 = 9.4 \text{ million}$$

其次，计算盈余的方差和波动率：

$$\text{Var}(\text{Surplus}) = (120 \times 0.12)^2 + (110 \times 0.04)^2 - 2 \times 0.7 \times (120 \times 0.12) \times (110 \times 0.04) = 137.916$$

$$\text{Volatility}(\text{Surplus}) = \sqrt{137.916} = 11.74 \text{ million}$$

最后，计算 95% 置信水平下的盈余下限：

$$\text{Lower Bound} = 9.4 - 1.645 \times 11.74 = 9.4 - 19.31 = -9.91 \text{ million}$$

- C 选项错误。**计算结果不符。
- D 选项错误。**计算结果不符。

考察: 养老金盈余风险价值计算

4 考点 4: 风险监测和绩效衡量

1. A portfolio manager holds 30,000 shares of Innovate Corp. in a portfolio. The average daily trading volume of Innovate Corp. is 80,000 shares. The manager wishes to trade no more than 10% of the daily trading volume of Innovate Corp. on any given day. Which of the following is closest to the liquidity duration of Innovate Corp. in this portfolio?
- A. 0.27
 - B. 0.375
 - C. 3.75
 - D. 25.0

答案:C

解析:

A 选项错误。计算不正确。

B 选项错误。计算时未考虑 10% 的交易限制。

C 选项正确。流动性久期 = 持有股数 / (每日最大交易比例 × 每日交易量) = 30,000 / (0.10 × 80,000) = 3.75 天。

D 选项错误。计算不正确。

考察: 流动性久期计算

5 考点 5: 投资组合绩效评估

1. Alice, a portfolio manager, claims her skill has consistently produced excess returns (over the benchmark) 99% of the time. To support this, she provides regression results from 60 monthly observations: $\alpha = 1.2\%$, and the standard error of $\alpha = 0.5\%$. Would you reject the null hypothesis of true $\alpha = 0$ and accept her claim of superior performance at a 99% confidence level?

- A. $t = 2.4$; fail to reject the null hypothesis; reject her claim.
- B. $t = 2.4$; reject the null hypothesis; accept her claim.
- C. $t = 1.96$; fail to reject the null hypothesis; accept her claim.
- D. $t = 1.96$; reject the null hypothesis; accept her claim.

答案:A

解析:

A 选项正确。 t 统计量 = α / α 的标准误 = $1.2\% / 0.5\% = 2.4$ 。在 99% 的置信水平下, 临界 t 值 (自由度约 60) 约为 2.66。由于 $2.4 < 2.66$, 我们无法拒绝真实 α 为 0 的原假设, 因此应拒绝她的说法。

B 选项错误。 t 统计量计算正确, 但结论错误。由于 t 值未超过 99% 置信水平的临界值, 不能拒绝原假设。

C 选项错误。 t 统计量计算错误, 1.96 是 95% 置信水平的临界值。

D 选项错误。 t 统计量计算错误, 并且结论也是错误的。

考察: Alpha 显著性的 t 检验

2. Assume a hedge fund reports a substantial positive alpha. The fund is known to use leverage and take both long and short positions in equities. The market experienced a downturn over the reporting period. Based on this information, which statement is most accurate?

- A. If the fund has a net positive beta, a portion of the alpha is attributable to the market’s performance.
- B. If the fund has a net negative beta, the alpha must entirely come from the market’s performance.
- C. If the fund has a net positive beta, the alpha must be generated entirely by stock selection, overcoming the market drag.
- D. If the fund has a net negative beta, a portion of the alpha is attributable to the market’s performance.

答案:D

解析:

A 选项错误。 当市场下跌时, 一个正 beta 的投资组合会受到市场的负面影响。因此, 任何正 alpha 都不可能来自市场表现。

B 选项错误。 虽然负 beta 在下跌的市场中会贡献正回报, 但 alpha 也包含了基金经理的选股能力, 因此不能说 alpha 完全来自市场。

C 选项错误。 这个说法比较接近, 但最准确的描述是选项 D。虽然基金经理的 alpha 确实需要克服市场拖累, 但选项 D 直接描述了负 beta 情况下的市场贡献。

D 选项正确。 因为市场下跌, 一个净负 beta 的投资组合 (例如, 净空头敞口) 会从市场下跌中获益。因此, 其报告的正 alpha 中, 有一部分可以归因于市场效应。

考察: Alpha 归因与市场表现

3. Leo Chen is an investment analyst for a European pension fund considering adding a large-cap equity mutual fund to its asset mix. To assess these funds, Leo conducts a detailed quantitative analysis on four mutual funds that claim to be large-capitalization funds. The results are in Exhibits 1 and 2.

Exhibit 1: Style Analysis Results					
		Orion	Pegasus	Phoenix	Hydra
MSCI World Value (large-cap)		97%	20%	30%	60%
MSCI World Growth (large-cap)		1%	70%	10%	30%
MSCI EM Value (small-cap)		2%	5%	35%	10%
MSCI EM Growth (small-cap)		0%	5%	25%	0%
Total		100%	100%	100%	100%
R^2		98.5%	90.1%	82.3%	70.2%
Exhibit 2: Performance Measurement					
	Benchmark (Euro Stoxx 50)	Orion	Pegasus	Phoenix	Hydra
Annual return (gross)	5.0%	5.2%	6.0%	6.5%	7.5%
Sharpe ratio	0.35	0.38	0.45	0.41	0.44
Treynor ratio	0.28	0.30	0.33	0.31	0.40
Tracking error	–	7.0%	8.0%	9.0%	8.5%

Based on these results, Leo made several comments. Which statement is least likely to be correct?

- A. Orion is a passively managed fund.
- B. The pension fund should invest in the Pegasus fund because it has the highest Sharpe ratio.
- C. Phoenix’ s investment style has drifted to small-capitalization.
- D. Hydra has the highest Information Ratio.

答案:B

解析:

A 选项错误。 Orion 基金的 R 平方值高达 98.5%，且其投资高度集中于大盘价值股，表明该基金很可能是一只被动管理型基金，该评论很可能是正确的。

B 选项正确。 夏普比率衡量总风险调整后的收益，但在考虑将基金加入现有投资组合时，它不一定是最佳指标。仅因夏普比率最高而做出投资决策是不全面的，因此该投资建议最不可能是正确的。

C 选项错误。 Phoenix 基金声称是大盘股基金，但其在小盘股上的配置合计达到了 60%（35% + 25%），表明其投资风格已发生漂移，该评论很可能是正确的。

D 选项错误。 信息比率 = (基金收益 - 基准收益) / 跟踪误差。Hydra 的信息比率 = (7.5% - 5.0%) / 8.5% ≈ 0.294，是四只基金中最高的，该评论很可能是正确的。

考察: 基金风格分析与业绩评估